

TURBULENT PIPE FLOW ANALYZER AND ENERGY SPECTRUM VISUALIZER

An Interactive Engineering Tool for Turbulence Analysis

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➤ Objective

This project aims to create an interactive tool to analyze turbulent pipe flows and visualize turbulence energy distribution, combining engineering calculations with dynamic visualizations.

➤ Background

Turbulence is a complex fluid flow regime characterized by chaotic, stochastic property changes. It plays a major role in aerospace, mechanical, and energy systems. Predicting flow behavior, pressure losses, and energy cascades is essential in engineering design.

➤ Methods

The analysis is based on the core principles of fluid mechanics:

- Reynolds number calculation to classify flow regimes (laminar, transitional, turbulent)
- Friction factor determination using exact and empirical formulas (Swamee-Jain approximation for turbulence)
- Pressure drop estimation via Darcy-Weisbach equation
- Turbulent intensity and turbulent kinetic energy estimation for deeper characterization

The energy cascade is simulated following Kolmogorov's $-5/3$ law, representing how energy transfers across scales in turbulent flows.

➤ Technologies Used

- **Microsoft Excel:** Used for engineering calculations, dynamic dashboards, live-updating graphs, conditional formatting, and interactive controls (sliders).
- **MATLAB:** Used to generate a log-log plot of the turbulence energy spectrum based on Kolmogorov's theory.

➤ Results and Features

- Fully automated engineering calculator for turbulent flow analysis
- Live updating graphs for Pressure Drop vs Flow Rate, and Friction Factor vs Reynolds Number
- Conditional formatting for flow regime visualization (green/yellow/red)
- Interactive slider to control Flow Rate dynamically
- MATLAB visualization of the turbulence energy cascade

➤ Bonus Features

- Professional layout and styling
- Automatic linking between inputs, calculations, and graphs
- MATLAB-based energy cascade plotting with labeled -5/3 slope

➤ Future Work

- Extend the tool to non-circular ducts and complex geometries
- Add CFD (Computational Fluid Dynamics) simulation data comparison
- Integrate machine learning to predict flow regime transitions

➤ Equations Used

The following equations were implemented in the Excel dashboard and MATLAB script to analyze turbulent pipe flow:

◇ 1. Cross-Sectional Area of Pipe

$$A = \frac{\pi D^2}{4}$$

◇ 2. Average Flow Velocity

$$V = \frac{Q}{A}$$

◇ 3. Reynolds Number

$$Re = \frac{\rho V D}{\mu}$$

◇ 4. Flow Regime Classification

- Laminar: $Re < 2300$
- Transitional: $2300 \leq Re \leq 4000$
- Turbulent: $Re > 4000$

◇ 5. Friction Factor

- For laminar flow: $f = \frac{64}{Re}$
- For turbulent flow (Swamee-Jain approximation): $f = \frac{0.25}{\left[\log_{10} \left(\frac{\epsilon}{3.7D} + \frac{5.74}{Re^{0.9}} \right) \right]^2}$

◇ 6. Pressure Drop (Darcy-Weisbach Equation)

$$\Delta P = f \cdot \frac{L}{D} \cdot \frac{\rho V^2}{2}$$

◇ 7. Turbulent Intensity (empirical estimate)

$$I = 0.16 \cdot Re^{-\frac{1}{8}}$$

◇ 8. Turbulent Kinetic Energy

$$k = \frac{3}{2}(I \cdot V)^2$$

◇ 9. Kolmogorov's Energy Spectrum (MATLAB Visualization)

$$E(k) \propto k^{-\frac{5}{3}}$$